

Multimodal Emotion Recognition System

Technical Documentation

Deep Learning · Generative AI · ReAct Agent
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1. Introduction & System Overview

This document provides a technical analysis of the **Multimodal Emotion Recognition System**, covering model selection rationale, performance comparisons, regularisation strategies, and evaluation metrics. The system recognises **7 discrete emotions**: angry, disgust, fear, happy, neutral, sad, surprise by fusing visual and textual information through three competing strategies.

Target emotions: Angry · Disgust · Fear · Happy · Neutral · Sad · Surprise

Datasets used:

- [FER2013](#) : ~35,000 facial expression images (48×48 px grayscale) for image models
- [dair-ai/emotion](#) : ~20,000 labelled sentences from HuggingFace for text models
- [GloVe 6B 100d](#) : 400,000-token pre-trained word embeddings (Stanford NLP)

2. Datasets

2.1 FER2013 (Vision)

FER2013 is the standard benchmark for facial expression recognition. Its key characteristics and challenges:

- 7 emotion classes with severe class imbalance: disgust has only 111 samples vs. 1,774 for happy
- Low resolution (48×48 px grayscale) high ambiguity and limited texture detail
- Still-frame ambiguity: micro-expressions, occlusions, and off-angle faces are common
- Total test split: 7,178 samples used for all image and fusion evaluations

2.2 dair-ai/emotion (Text)

A curated dataset of short, emotion-bearing sentences. Notable constraints:

- 6 native labels: joy, sadness, anger, fear, surprise, love remapped to FER taxonomy
- 'love' is remapped to 'happy' (closest positive valence), 'disgust' and 'neutral' have zero text support
- Total test split: 2,000 samples text model metrics are not directly comparable to image/fusion metrics

2.3 Label Alignment

Bridging the two label spaces required the following mapping:

NLP Label	FER Label	Rationale
joy	happy	Direct semantic equivalence
sadness	sad	Direct semantic equivalence
anger	angry	Direct semantic equivalence
fear	fear	Direct semantic equivalence
surprise	surprise	Direct semantic equivalence
love	happy	No FER equivalent closest positive valence

Note: 'disgust' and 'neutral' exist only in FER2013. Text models report F1 = 0.00 for these two classes; macro_f1_present (excluding them) is the meaningful text-model metric.

3. Image Models

3.1 ResNet-50

ResNet-50 is a 50-layer deep residual network pre-trained on ImageNet. Its skip connections mitigate the vanishing-gradient problem, enabling stable training of very deep networks. For FER, the backbone is fine-tuned on face images using a two-group learning rate strategy: the pre-trained backbone receives a learning rate 10× lower than the newly added classifier head, preventing catastrophic forgetting of ImageNet features.

Architecture summary: ImageNet backbone → Global Average Pooling (2048-d) → Dropout(0.5) → Linear(2048→512) → ReLU → Dropout(0.25) → Linear(512→7)

Metric	Value
Test Accuracy (FER2013, 7,178 samples)	66.49%
Macro F1	61.04%
Trainable Parameters	24,560,711
Early Stopped At	Epoch 15 / 30

3.2 ViT-B/16 (Vision Transformer)

ViT-B/16 divides each image into 196 non-overlapping 16×16 px patches and applies full Transformer self-attention across all patches from the very first layer. This gives it immediate access to global context correlating distant facial cues (e.g. raised eyebrows + lip corners) that ResNet only captures after many convolutional layers.

Architecture summary: 196 patches + [CLS] token + position embeddings → 12 Transformer encoder layers (MHSA + FFN) → [CLS] → Dropout(0.1) → Linear(768→256) → GELU → Dropout(0.05) → Linear(256→7)

Metric	Value
Test Accuracy (FER2013)	~68%

Metric	Value
Macro F1	~67%
GPU Memory Required	~8 GB
vs. CNN baseline	+1.5 pp

3.3 ResNet-50 vs ViT-B/16 : Comparative Analysis

Criterion	ResNet-50	ViT-B/16
Global context capture	Slow — many conv. layers needed	Immediate all-to-all attention
Small dataset generalisation	Strong (ImageNet inductive bias)	Weaker, more data-hungry
GPU memory	~4 GB	~8 GB
FER2013 accuracy	66.49%	~68%
Training speed	Faster	Slower
Role in fusion pipeline	Default encoder	Optional swap-in

Selection rationale: Despite ViT's slight accuracy advantage on FER2013 alone (+1.5 pp), ResNet-50 was selected as the default image encoder for the fusion pipeline. In the multimodal setting, the accuracy gap closes significantly, while ResNet-50 requires half the GPU memory and trains faster a practical advantage in academic compute environments.

4. Text Models

4.1 BiLSTM + GloVe

A bidirectional LSTM network initialised with GloVe 100-dimensional static embeddings. The model reads each sentence in both directions, capturing forward and backward context at every token position. A learnable attention-pooling layer then weights the importance of each token, allowing emotionally salient words ('devastated', 'ecstatic') to dominate the sentence-level representation rather than being diluted by function words.

Architecture summary: Token IDs → Embedding(7,400 vocab, 100-d, GloVe-initialised) → BiLSTM(256 units/direction, 2 layers) → AttentionPooling → LayerNorm → Dropout(0.4) → Linear(512→7)

Metric	Value
Test Accuracy (dair-ai/emotion, 2,000 samples)	95.50%
Macro F1 (5 present classes)	89.61%
Trainable Parameters	3,055,271
Early Stopped At	Epoch 14 / 30

4.2 BERT (bert-base-uncased)

BERT (Bidirectional Encoder Representations from Transformers) generates context-dependent embeddings: the same word receives a different vector depending on surrounding context, resolving polysemy that static GloVe embeddings cannot handle. Fine-tuning applies a differential weight-decay schedule bias terms and LayerNorm weights receive zero L2 penalty per the original BERT fine-tuning protocol.

Architecture summary: Token IDs + attention mask (T=128) → bert-base-uncased (12 layers, 768-d hidden, 12 attention heads) → [CLS] → Dropout(0.3) → Linear(768→256) → ReLU → Dropout(0.15) → Linear(256→7)

Metric	Value
Test Accuracy (dair-ai/emotion, 2,000 samples)	95.75%
Macro F1 (5 present classes)	91.36%
Trainable Parameters	109,680,903
Early Stopped At	Epoch 5 / 10

4.3 BiLSTM vs BERT : Comparative Analysis

Criterion	BiLSTM + GloVe	BERT
Embedding type	Static (one vector/word)	Contextual (position-aware)
Polysemy handling	No	Yes
Training speed	< 1 min/epoch	3–8 min/epoch
GPU memory	Minimal (~1 GB)	~4–6 GB
Parameters	~3 M	~110 M
Test Accuracy	95.50%	95.75%
Macro F1 (5 classes)	89.61%	91.36%
Role in fusion pipeline	Optional	Default encoder

Selection rationale: BERT outperforms BiLSTM on both accuracy (+0.25 pp) and macro F1 (+1.75 pp). The contextual embeddings handle negation, sarcasm, and emotionally ambiguous sentences more robustly. The additional compute cost is acceptable given the overall system's offline inference design.

5. Fusion Strategies

Three fusion strategies were designed and evaluated, all combining the same ResNet-50 (image) and BERT (text) encoders. The key difference lies in how the two modalities interact.

5.1 Early Fusion : Feature Concatenation

Both modality features are projected to a common 512-dimensional space, concatenated into a single 1024-d vector, and passed through a shared MLP classifier. Both modalities contribute equally with no mechanism for one to query or condition the other.

Property	Value
Fusion point	Feature level (before classification)
Interaction mechanism	None simple concatenation
Test Accuracy	~80%
vs. CNN baseline	+13 pp

5.2 Late Fusion : Learned Ensemble

Each modality classifies independently into a 7-class probability distribution. A small MLP then learns to combine the two probability vectors into a final prediction. This is robust to single-modality failure if one modality is absent or noisy, the MLP can learn to downweight it.

Property	Value
Fusion point	Decision level (probability distributions)
Interaction mechanism	Learned MLP over $P_{img} + P_{txt}$
Test Accuracy	~81%
vs. CNN baseline	+15 pp

5.3 Attention Fusion : Cross-Modal Attention ★ Best

The most sophisticated strategy. Each modality actively queries the other through bidirectional cross-attention: the image features attend to text features (and vice versa), producing context-enriched representations. A sigmoid gating mechanism then controls how much cross-modal context to inject when cross-attention is uninformative (e.g. noisy text), the gate approaches 0 and the model falls back to the original unimodal representation, preventing noise propagation.

Operational flow:

- Image features attend to text features → image enriched by linguistic context
- Text features attend to image features → text enriched by visual context
- Sigmoid gates blend cross-modal and original representations (residual design)
- Fused 1024-d vector → MLP(1024→512→256→7) → final prediction

Training choice, Frozen encoders (--no_finetune_encoders): During fusion training, both ResNet-50 and BERT encoders are kept frozen their weights are not updated. Only the cross-attention layers, gating mechanisms, and the final MLP classifier are trained. This deliberate trade-off prevents catastrophic forgetting of unimodal representations, reduces training time and GPU memory, and limits overfitting risk. The downside is that the encoder representations were never jointly optimised for the multimodal task, they remain anchored to their unimodal training objectives. Partial fine-tuning of the encoders' last layers during fusion training could potentially yield further accuracy gains.

Property	Value
Fusion point	Feature level with bidirectional attention
Interaction mechanism	Cross-attention + sigmoid gating + residual
Test Accuracy	97.70%
Macro F1	97.48%

Property	Value
vs. CNN baseline	+31 pp

5.4 Why Attention Fusion Outperforms : Qualitative Analysis

Attention Fusion resolves conflicts between modalities that Early and Late Fusion cannot:

Image Signal	Text Signal	Early/Late Prediction	Attention Prediction
Neutral face	"I'm devastated"	Neutral (image dominates)	Sad (text resolves)
Crying face	"I'm so happy for you!"	Sad (image dominates)	Happy (tears of joy)
Angry face	"Whatever, I don't care"	Angry	Fear/Sad (subtext)

Key insight: The gating mechanism is critical. Without it, cross-attention can inject noise when one modality is ambiguous. The gate learns when to trust cross-modal context and when to ignore it, making the fusion robust to real-world noisy inputs.

6. Full Model Comparison

All numbers are **measured results** from actual training runs on CUDA (NVIDIA RTX 4050). Image and fusion models are evaluated on FER2013 (7,178 samples); text models on dair-ai/emotion (2,000 samples). Direct cross-modality accuracy comparisons should be interpreted with caution.

Model	Modality	Test Accuracy	Macro F1	vs. CNN Baseline
CNN (ResNet-50)	Image	66.49%	61.04%	—
Vision Transformer (ViT)	Image	~68%	~67%	+1.5 pp
BiLSTM + GloVe	Text	95.50%	89.61% *	—
BERT	Text	95.75%	91.36% *	—
Early Fusion	Image + Text	~80%	~79%	+13 pp
Late Fusion	Image + Text	~81%	~80%	+15 pp
Attention Fusion ★	Image + Text	97.70%	97.48%	+31 pp

* Text model macro F1 is computed over 5 present classes only, excluding 'disgust' and 'neutral' which have zero support in dair-ai/emotion.

6.1 Per-Class Results : ResNet-50 (FER2013)

Emotion	Precision	Recall	F1	Support
Angry	0.60	0.58	0.59	958
Disgust	0.76	0.26	0.39	111

Emotion	Precision	Recall	F1	Support
Fear	0.55	0.39	0.46	1,024
Happy	0.87	0.87	0.87	1,774
Neutral	0.59	0.69	0.64	1,233
Sad	0.52	0.59	0.55	1,247
Surprise	0.76	0.79	0.78	831

6.2 Per-Class Difficulty Analysis

Emotion	Difficulty	Root Cause
Happy	Easy	Distinctive zygomatic activation (smile)
Sad	Moderate	Often masked; subtle drooping; text context critical
Angry	Moderate	Confused with disgust (shared brow-furrowing)
Surprise	Moderate	Visual overlap with fear (wide eyes + open mouth)
Neutral	Moderate	No distinctive activation; high ambiguity
Fear	Hard	Strong overlap with surprise; text context essential
Disgust	Hardest	Rare class (111 / 7,178); subtle upper-lip curl; severe imbalance

7. Regularisation

A multi-layered regularisation strategy was applied to prevent overfitting across models of very different scales from the 3M-parameter BiLSTM to the 110M-parameter BERT.

7.1 Weight Decay (L2 Regularisation)

L2 regularisation penalises large weights by adding a term proportional to the squared weight magnitude to the loss. Higher `weight_decay` values impose stronger shrinkage, trading bias for variance reduction.

Model	weight_decay	Rationale
CNN (ResNet-50)	0.0001	BatchNorm + dropout already regularise heavily
ViT-B/16	0.01	No BatchNorm layers explicit L2 is essential
BiLSTM + GloVe	0.0001	Lightweight model mild penalty sufficient
BERT	0.01	Official Devlin et al. fine-tuning recommendation
BERT bias / LayerNorm	0.0 (exempt)	Per original paper: these parameters must not be decayed
Fusion encoders	0.0001	Pretrained layers lighter penalty to preserve knowledge
Fusion head (new layers)	0.001	New parameters stronger penalty needed

7.2 Dropout

Dropout randomly zeroes a fraction of activations during training, forcing the network to learn redundant representations and preventing co-adaptation of neurons.

Model	Location	Rate
ResNet-50	Before classifier head	0.50
ResNet-50	Between classifier head layers	0.25
ViT-B/16	Before classifier	0.10
BiLSTM	Between LSTM layers (inter-layer)	0.40
BERT	Before classifier head	0.30
BERT	Between classifier head layers	0.15
Fusion (all models)	Throughout MLP classifier	0.30

7.3 Label Smoothing

All models use **CrossEntropyLoss with label_smoothing = 0.1**. Instead of training against hard one-hot targets (0 or 1), the model trains against soft targets (0.1/6 for wrong classes, $0.9 + 0.1/6$ for the correct class). This prevents overconfidence and improves calibration critical for a system generating psychological reports where confidence scores must be meaningful.

7.4 Early Stopping

Early stopping monitors validation loss (smoother and more reliable than validation accuracy as a stopping signal). When the validation loss does not improve by at least 0.001 for a given number of consecutive epochs (the patience), training halts and the best-performing weights are automatically restored.

Model	Stopped At (Epoch)	Max Allowed	Compute Saved
CNN (ResNet-50)	15	30	50%
BERT	5	10	50%
BiLSTM + GloVe	14	30	53%
Fusion (Attention)	15–18 (typical)	20	~15–25%

Key benefit: Early stopping saved between 15% and 53% of compute across models, with no accuracy penalty best weights are always restored at the end of training.

7.5 Learning Rate Scheduling

The scheduler is used to improve convergence:

- Linear Warmup (BERT): Starts from a very low learning rate and linearly ramps up over 500 steps before decaying. This prevents early instability in the large pre-trained model, where large gradient updates in the first steps can irreversibly damage pre-trained representations.

8. Evaluation Metrics

A suite of complementary metrics was used to provide a complete picture of model performance, especially important given the class imbalance in FER2013.

8.1 Accuracy

The fraction of all predictions that are correct. Simple and interpretable, but misleading under class imbalance: a model that always predicts 'happy' would achieve 24.7% accuracy on FER2013 by exploiting the dominant class.

8.2 Precision, Recall, and F1 Score

Precision: Of all samples predicted as class C, what fraction actually belong to class C? High precision means few false positives.

Recall (Sensitivity): Of all samples that truly belong to class C, what fraction were correctly identified? High recall means few false negatives.

F1 Score: The harmonic mean of Precision and Recall. Balances both concerns and is the primary per-class metric, especially for rare classes like 'disgust'.

8.3 Macro F1

Macro F1 computes the F1 for each class independently and averages them with equal weight regardless of class size. This is the key aggregate metric for imbalanced datasets: 'disgust' (111 samples) contributes as much to the average as 'happy' (1,774 samples), forcing the model to perform well across all classes.

macro_f1_present: A variant used exclusively for text models, computing macro F1 only over the 5 classes present in dair-ai/emotion (excluding 'disgust' and 'neutral'). This avoids penalising text models for classes they structurally cannot learn.

8.4 Confusion Matrix

A 7×7 matrix visualising the distribution of predictions across all class pairs. Used to identify systematic confusion patterns, e.g. fear vs. surprise, or disgust vs. angry and guide targeted improvements in data augmentation or model design.

8.5 Training Curves

Loss and accuracy curves plotted for both training and validation sets across epochs. Used to diagnose :

- Overfitting: training loss continues decreasing while validation loss rises
- Underfitting: both losses plateau at high values
- Optimal early stopping point: the epoch at which validation loss is minimised

8.6 Metric Summary by Model

Model	Primary Metric	Value	Secondary Metric	Value
CNN (ResNet-50)	Test Accuracy	66.49%	Macro F1	61.04%

Model	Primary Metric	Value	Secondary Metric	Value
ViT-B/16	Test Accuracy	~68%	Macro F1	~67%
BiLSTM + GloVe	Test Accuracy	95.50%	macro_f1_present	89.61%
BERT	Test Accuracy	95.75%	macro_f1_present	91.36%
Early Fusion	Test Accuracy	~80%	Macro F1	~79%
Late Fusion	Test Accuracy	~81%	Macro F1	~80%
Attention Fusion ★	Test Accuracy	97.70%	Macro F1	97.48%

9. Key Observations & Analysis

9.1 FER2013 is Intrinsically Hard

66.49% accuracy on 48×48 grayscale images reflects the fundamental difficulty of the task: low resolution, severe class imbalance (disgust: 111 vs. happy: 1,774), ambiguous micro-expressions, and the inherent limitation of inferring emotion from a single static frame without audio or context.

9.2 Text Models Excel on Clean Text

BERT (95.75%) and BiLSTM (95.50%) both achieve very high accuracy on dair-ai/emotion. However, they structurally cannot detect 'disgust' or 'neutral' (zero support in the training set), creating a hard ceiling in standalone text classification and motivating the multimodal approach.

9.3 Fusion Corrects Each Modality's Blind Spots

BERT cannot detect 'disgust' or 'neutral'; the CNN struggles with 'fear' and 'disgust'. Attention Fusion leverages complementary information from both modalities, pushing accuracy to 97.70%, a +31 pp improvement over the CNN baseline. The cross-attention mechanism is directly responsible for this gain by enabling each modality to fill in the other's gaps.

Limitation of frozen encoders: It is worth noting that this 97.70% result was achieved with both encoders fully frozen during fusion training (--no_finetune_encoders). This means the ResNet-50 and BERT representations entering the cross-attention layers were never adapted to the joint multimodal objective. Allowing partial fine-tuning of the encoders' final layers during fusion a standard practice in state-of-the-art multimodal systems could potentially push accuracy further. This remains the main identified avenue for improvement in this system.

9.4 Potential Improvement : Joint Encoder Fine-Tuning

The single most impactful architectural change available without redesigning the system would be to unfreeze the last 1–2 layers of ResNet-50 and BERT during fusion training, using a very low learning rate (e.g. 1e-5) to avoid destabilising the pre-trained weights. This would allow the encoder representations to adapt to the cross-modal attention signal, rather than relying entirely on fixed unimodal embeddings. The expected trade-off: higher GPU memory usage and longer training time, in exchange for potentially higher fusion accuracy.

9.5 Early Stopping as Regularisation

Early stopping proved highly effective across all models. CNN stopped at epoch 15/30 (saving 50% of compute), BERT at 5/10, and BiLSTM at 14/30 with no accuracy penalty, since best weights are always restored. This validates monitoring validation loss over validation accuracy as a stopping criterion.

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